



UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO
FACULTAD DE CIENCIAS



Fundamentos de Bases de Datos

Bitácora Instalación Docker y del SMBD, PostgreSQL.

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Grupo: 7112

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- **Sistema Operativo:** GNU/Linux
- **Versión:** 40
- **Distribución:** Fedora Linux (Workstation Edition)
- **Versión de la instalación:**
 - **Docker:** version 28.1.1, build 4eba377
 - **PostgreSQL:** 18.6 (Debian 18.6-1.pgdg13+2)
 - **DBeaver Community:** 26.2.0.202608301738
- **Tiempo requerido:** 65 minutos

1. Instalación Docker

1. Actualizamos la lista de paquetes existentes:

```
sudo dnf check-update
```

```
~ » sudo dnf check-update
[sudo] contraseña para alexareynoso:
Copr repo for PyCharm owned by phracek
Errors during downloading metadata for repository 'copr:copr.fedorainfracloud.org:phracek:PyCharm':
- Status code: 404 for https://download.copr.fedorainfracloud.org/results/phracek/PyCharm/fedora-40-x86_64/repodata/repomd.xml (IP: 2600:9000:21d5:9600:4:bbc1:1840:93a1)
Error: Failed to download metadata for repo 'copr:copr.fedorainfracloud.org:phracek:PyCharm': Cannot download repomd.xml: Cannot download repodata/repomd.xml: All mirrors were tried
Descartando repositorios: copr:copr.fedorainfracloud.org:phracek:PyCharm
Última comprobación de caducidad de metadatos hecha hace 4:25:48, el sáb 05 sep 2026 15:16:52.
```

2. Instalamos los paquetes necesarios para agregar el repositorio de Docker:

```
sudo dnf -y install dnf-plugins-core
```

```
~ » sudo dnf -y install dnf-plugins-core
Copr repo for PyCharm owned by phracek
Errors during downloading metadata for repository 'copr:copr.fedorainfracloud.org:phracek:PyCharm':
- Status code: 404 for https://download.copr.fedorainfracloud.org/results/phracek/PyCharm/fedora-40-x86_64/repodata/repomd.xml (IP: 2600:9000:21d5:ea00:4:bbc1:1840:93a1)
Error: Failed to download metadata for repo 'copr:copr.fedorainfracloud.org:phracek:PyCharm': Cannot download repomd.xml: Cannot download repodata/repomd.xml: All mirrors were tried
Descartando repositorios: copr:copr.fedorainfracloud.org:phracek:PyCharm
Última comprobación de caducidad de metadatos hecha hace 4:28:57, el sáb 05 sep 2026 15:16:52.
El paquete dnf-plugins-core-4.10.1-1.fc40.noarch ya está instalado.
Dependencias resueltas.
Nada por hacer.
¡Listo!
```

3. Descargamos la llave GPG oficial de Docker:

```
sudo dnf config-manager --add-repo https://download.docker.com/linux/fedora/docker-ce.repo
```

```
~ » sudo dnf config-manager --add-repo https://download.docker.com/linux/fedora/docker-ce.repo
Agregando repositorio de: https://download.docker.com/linux/fedora/docker-ce.repo
```

4. Instalar Docker y sus complementos:

```
sudo dnf install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

```
alexareynoso@fedora ~
- > sudo dnf install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
[sudo] contraseña para alexareynoso:
Copr repo for PyCharm owned by phracek                                146 B/s | 158 B   00:01
Errors during downloading metadata for repository 'copr:copr.fedorainfracloud.org:phracek:PyCharm':
- Status code: 404 for https://download.copr.fedorainfracloud.org/results/phracek/PyCharm/fedora-40-x86_64/repodata/repomd.xml (IP: 2600:9000:21d5:e800:4:bbc1:1840:93a1)
Error: Failed to download metadata for repo 'copr:copr.fedorainfracloud.org:phracek:PyCharm': Cannot download repomd.xml: Cannot download repodata/repomd.xml: All mirrors were tried
Docker CE Stable - x86_64                                           78 kB/s | 17 kB   00:00
Descartando repositorios: copr:copr.fedorainfracloud.org:phracek:PyCharm
Dependencias resueltas.
=====
Paquete                    Arquitectura      Versión          Repositorio          Tam.
=====
Instalando:
containerd.io              x86_64           1.7.27-3.1.fc40  docker-ce-stable     44 M
docker-buildx-plugin       x86_64           0.23.0-1.fc40    docker-ce-stable     16 M
docker-ce                  x86_64           3:28.1.1-1.fc40  docker-ce-stable     20 M
docker-ce-cli              x86_64           1:28.1.1-1.fc40  docker-ce-stable     8.3 M
docker-compose-plugin      x86_64           2.35.1-1.fc40    docker-ce-stable     15 M
Instalando dependencias:
libcgroup                  x86_64           3.0-5.fc40       fedora                74 k
slirp4netns                x86_64           1.3.1-1.fc40     updates              47 k
Instalando dependencias débiles:
docker-ce-rootless-extras x86_64           28.1.1-1.fc40    docker-ce-stable     3.2 M

Resumen de la transacción
=====
Instalar 8 Paquetes

Tamaño total de la descarga: 106 M
Tamaño instalado: 426 M
¿Está de acuerdo [s/N]? s
Descargando paquetes:
(1/8): docker-buildx-plugin-0.23.0-1.fc40.x86_64.rpm               4.7 MB/s | 16 MB   00:03
(2/8): docker-ce-28.1.1-1.fc40.x86_64.rpm                       3.6 MB/s | 20 MB   00:05
(3/8): docker-ce-cli-28.1.1-1.fc40.x86_64.rpm                   2.7 MB/s | 8.3 MB   00:03
(4/8): docker-ce-rootless-extras-28.1.1-1.fc40.x86_64.rpm       1.9 MB/s | 3.2 MB   00:01
(5/8): libcgroup-3.0-5.fc40.x86_64.rpm                           73 kB/s | 74 kB   00:01

(6/8): slirp4netns-1.3.1-1.fc40.x86_64.rpm                       202 kB/s | 47 kB   00:00
(7/8): containerd.io-1.7.27-3.1.fc40.x86_64.rpm                 4.5 MB/s | 44 MB   00:09
(8/8): docker-compose-plugin-2.35.1-1.fc40.x86_64.rpm           3.8 MB/s | 15 MB   00:03

-----
Total                                                                9.7 MB/s | 106 MB   00:10
Docker CE Stable - x86_64                                           28 kB/s | 1.6 kB   00:00
Importando llave GPG 0x621E9F35:
ID usuario: "Docker Release (CE rpm) <docker@docker.com>"
Huella   : 060A 61C5 1B55 8A7F 742B 77AA C52F EB6B 621E 9F35
Desde    : https://download.docker.com/linux/fedora/gpg
¿Está de acuerdo [s/N]? s
La llave ha sido importada exitosamente
Ejecutando verificación de operación
Verificación de operación exitosa.
Ejecutando prueba de operaciones
Prueba de operación exitosa.
Ejecutando operación
Preparando :                                                                1/1
Instalando : docker-buildx-plugin-0.23.0-1.fc40.x86_64                  1/8
Ejecutando scriptlet: docker-buildx-plugin-0.23.0-1.fc40.x86_64        1/8
Instalando : docker-compose-plugin-2.35.1-1.fc40.x86_64                2/8
Ejecutando scriptlet: docker-compose-plugin-2.35.1-1.fc40.x86_64        2/8
Instalando : docker-ce-cli-1:28.1.1-1.fc40.x86_64                     3/8
Ejecutando scriptlet: docker-ce-cli-1:28.1.1-1.fc40.x86_64             3/8
Instalando : slirp4netns-1.3.1-1.fc40.x86_64                          4/8
Instalando : libcgroup-3.0-5.fc40.x86_64                              5/8
Instalando : containerd.io-1.7.27-3.1.fc40.x86_64                    6/8
Ejecutando scriptlet: containerd.io-1.7.27-3.1.fc40.x86_64            6/8
Instalando : docker-ce-rootless-extras-28.1.1-1.fc40.x86_64          7/8
Ejecutando scriptlet: docker-ce-rootless-extras-28.1.1-1.fc40.x86_64  7/8
Instalando : docker-ce-3:28.1.1-1.fc40.x86_64                        8/8
Ejecutando scriptlet: docker-ce-3:28.1.1-1.fc40.x86_64                8/8

Instalado:
containerd.io-1.7.27-3.1.fc40.x86_64  docker-buildx-plugin-0.23.0-1.fc40.x86_64  docker-ce-3:28.1.1-1.fc40.x86_64
docker-ce-cli-1:28.1.1-1.fc40.x86_64  docker-ce-rootless-extras-28.1.1-1.fc40.x86_64  docker-compose-plugin-2.35.1-1.fc40.x86_64
libcgroup-3.0-5.fc40.x86_64          slirp4netns-1.3.1-1.fc40.x86_64

¡Listo!
```

5. Comprobar e iniciar el servicio:

```
sudo systemctl enable --now docker
sudo systemctl status docker
```

```
CPU: 731ms
CGroup: /system.slice/docker.service
└─15550 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

sep 05 20:09:02 fedora dockerd[15550]: time="2026-09-05T20:09:02.957783712-06:00" level=info msg="Creating a containerd client" address=/run/
sep 05 20:09:03 fedora dockerd[15550]: time="2026-09-05T20:09:03.107389887-06:00" level=info msg="Loading containers: start."
sep 05 20:09:03 fedora dockerd[15550]: time="2026-09-05T20:09:03.175288716-06:00" level=info msg="Firewalld: created docker-forwarding poli
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.678684512-06:00" level=info msg="Loading containers: done."
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.732332968-06:00" level=info msg="Docker daemon" commit=01f442b containerd=
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.732527584-06:00" level=info msg="Initializing buildkit"
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.975752396-06:00" level=info msg="Completed buildkit initialization"
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.998752924-06:00" level=info msg="Daemon has completed initialization"
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.998980060-06:00" level=info msg="API listen on /run/docker.sock"
sep 05 20:09:06 fedora systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-24/24 (END) ...skipping...
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/service.d
            └─10-timeout-abort.conf
   Active: active (running) since Sat 2026-09-05 20:09:06 CST; 211ms ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 15550 (dockerd)
      Tasks: 14
     Memory: 62.7M (peak: 65.4M)
        CPU: 731ms
     CGroup: /system.slice/docker.service
             └─15550 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

sep 05 20:09:02 fedora dockerd[15550]: time="2026-09-05T20:09:02.957783712-06:00" level=info msg="Creating a containerd client" address=/run/
/containers/containerd.sock timeout=1m0s
sep 05 20:09:03 fedora dockerd[15550]: time="2026-09-05T20:09:03.107389887-06:00" level=info msg="Loading containers: start."
sep 05 20:09:03 fedora dockerd[15550]: time="2026-09-05T20:09:03.175288716-06:00" level=info msg="Firewalld: created docker-forwarding polic
y"
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.678684512-06:00" level=info msg="Loading containers: done."
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.732332968-06:00" level=info msg="Docker daemon" commit=01f442b containerd=s
napshotter=false storage-driver=overlay2 version=28.1.1
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.732527584-06:00" level=info msg="Initializing buildkit"
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.975752396-06:00" level=info msg="Completed buildkit initialization"
sep 05 20:09:05 fedora dockerd[15550]: time="2026-09-05T20:09:05.998752924-06:00" level=info msg="Daemon has completed initialization"
```

6. Probamos el funcionamiento con una imagen de prueba:

```
sudo docker run hello-world
```

```
alexareynoso@fedora
~ » sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

2. Instalación Docker

1. Descargar la imagen oficial de PostgreSQL:

```
sudo docker pull postgres
```

```
~ » sudo docker pull postgres
[sudo] contraseña para alexareynoso:
Using default tag: latest
latest: Pulling from library/postgres
6310eb16bf42: Pull complete
6463c6b5c703: Pull complete
8d6422da287d: Pull complete
2fc0e69f8d7d: Pull complete
5fa29c524b4e: Pull complete
ace4b5b96a91: Pull complete
af2a6480a058: Pull complete
5ee6c6ca7fd9: Pull complete
c5694bff1357: Pull complete
aedab7177504: Pull complete
a4635eae9b8b: Pull complete
46e663abc8f5: Pull complete
1bb54e8f7727: Pull complete
Digest: sha256:4ef4dbc939d61acea57712655ddb4b4ab27419c913f94cca0cd57cb3ea3c2280
Status: Downloaded newer image for postgres:latest
docker.io/library/postgres:latest
```

2. Crear e iniciar el contenedor de PostgreSQL

```
sudo docker run -d
--name postgres
-e POSTGRES_PASSWORD=mysecretpassword
-p 127.0.0.1:5432:5432
-v postgres_data:/var/lib/postgresql
postgres
```

3. Obtenemos su ID y probamos detenerlo e iniciarlo con:

```
sudo docker ps -a
```

```
~ » sudo docker ps -a
[sudo] contraseña para alexareynoso:
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
c4ed27ead21f   postgres      "docker-entrypoint.s..." 5 minutes ago  Up 5 minutes  127.0.0.1:5432->5432/tcp  postgres
fc0da1ea12af   hello-world    "/hello"                 14 minutes ago Exited (0) 14 minutes ago  sad_banzai
```

4. Obtenemos su ID y probamos detenerlo e iniciarlo con:

```
sudo docker ps -a
```

```
~ » sudo docker ps -a
[sudo] contraseña para alexareynoso:
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
c4ed27ead21f   postgres      "docker-entrypoint.s..." 5 minutes ago  Up 5 minutes  127.0.0.1:5432->5432/tcp  postgres
fc0da1ea12af   hello-world    "/hello"                 14 minutes ago Exited (0) 14 minutes ago  sad_banzai
```

```
sudo docker stop postgres
sudo docker start postgres
```

```
~ » sudo docker stop postgres
postgres
~ » sudo docker start postgres
postgres
```

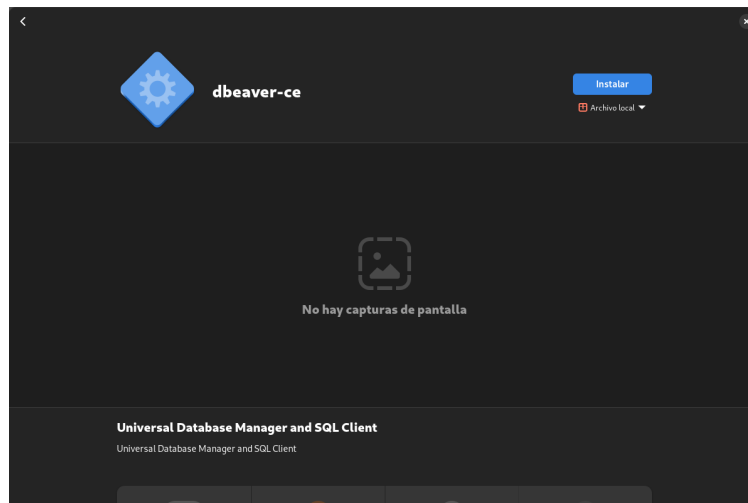
5. Conectamos a PostgreSQL una vez creado el contenedor podemos utilizar psql:

```
sudo docker run -it --link postgres:postgres postgres psql -h postgres -U
postgres
```

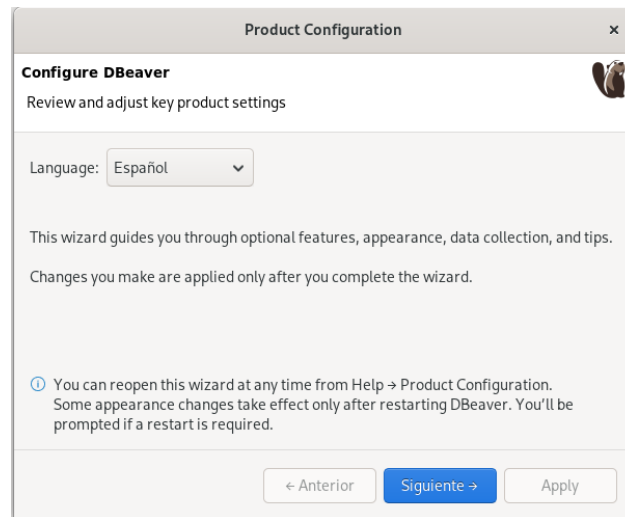
```
~ » sudo docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
Password for user postgres:
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.
postgres=# \q
```

3. DBeaver

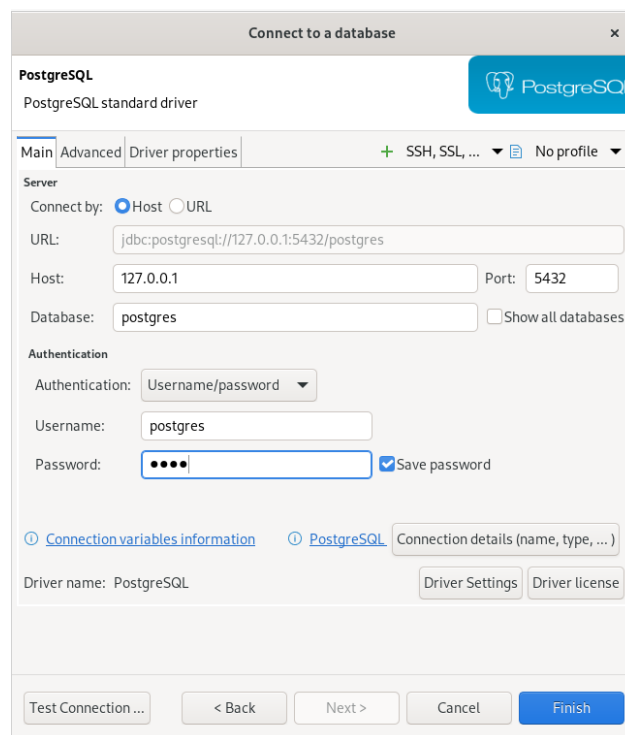
1. Instalamos Dbeaver mediante el enlace de descarga:



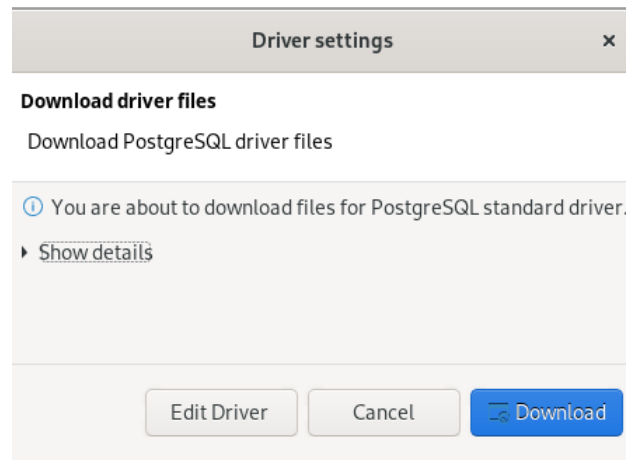
2. Configuramos DBeaver:



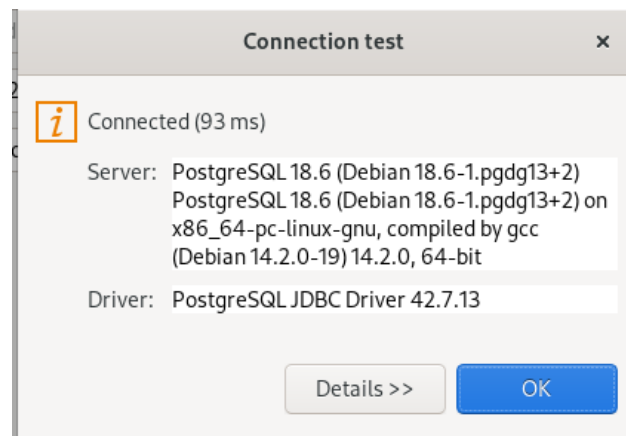
3. Conectamos DBeaver:



Nos pide descargar unos drives



Verificamos que la conexión sea exitosa



4. Comentario

Durante la instalación de los programas no tuve ningún problema o error que me impidiera descargarlos; incluso para conectar PostgreSQL a DBeaver fue bastante sencillo. El único inconveniente fue tener que buscar y adaptar los comandos adecuados para mi distribución (Fedora Linux), cambiando apt por dnf para descargar Docker y DBeaver.

A pesar de ese detalle con la distribución, las instrucciones fueron bastante claras para instalar todo, correr el contenedor de PostgreSQL y probar la conexión. Por lo que la práctica fue bastante útil para entender qué es y qué hace cada uno de ellos, y así poder desarrollar las prácticas futuras a más detalle.